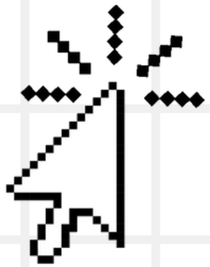
FPT UNIVERSITY

Vietnamese Legal Assistant: Multi-agent Collaborative System using Knowledge Graph and Large Language Model



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Ta Van Dai
Supervisor: Assoc. Prof. Phan Duy Hung

AIP490_G8 |
Fall_2024

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1 INTRODUCTION

MOTIVATION

- 38,371 legal aid cases [1]
- Vietnamese Legal Dictionary
- Many countries have legal assistant chatbots



RELATED WORKS

Benchmark for assessing different LLM capabilities

Benchmark	Evaluation Test
AGIEval	Human-centric standardized tests
HELM	Sixteen core scenario
KOLA	Knowledge-oriented assessment
MMBench	Vision-language models

RELATED WORKS

Legal Benchmark

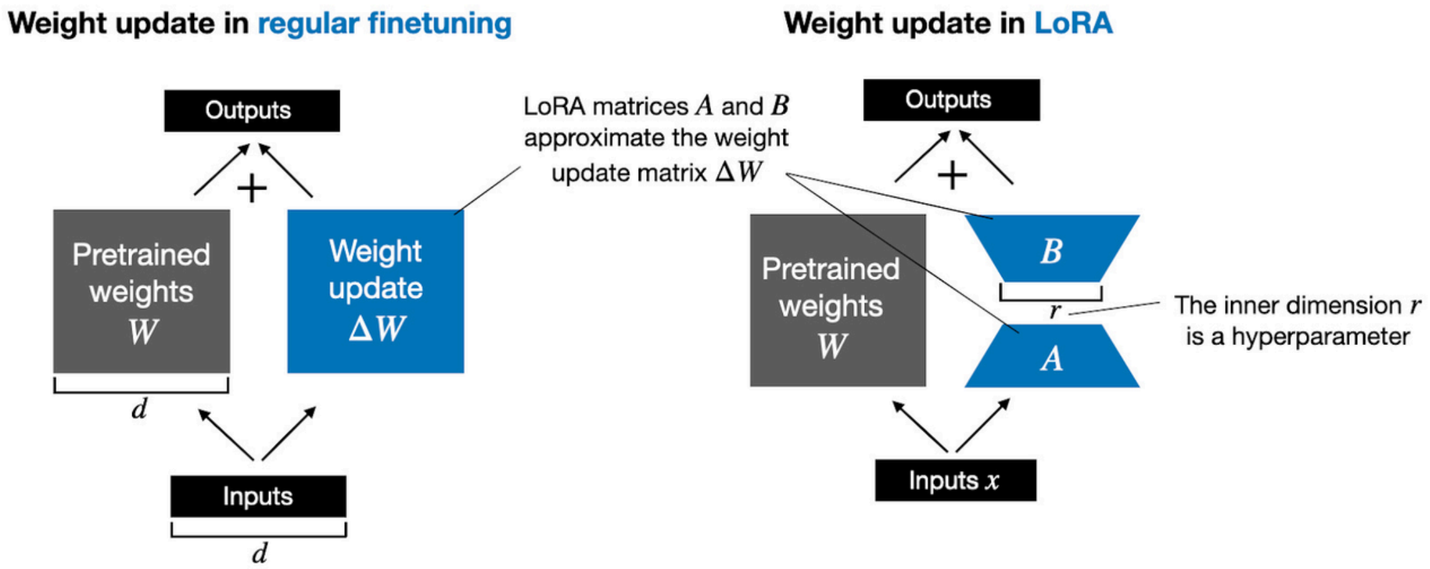
- In many countries, there are several legal benchmark for evaluating the performance of Large Language Models (LLM), here are some remarkable example

Benchmark	Nation
Lexglue	US and EU
LBOX OPEN	South Korean
LawBench	China
Legal Bench	US

RELATED WORKS

Finetuning LLM

Low-Rank Adaptation of Large Language Models (LoRA)



RELATED WORKS

Legal Question Answering

- **Domain-specific large language models**
  - Domain-adaption: biology, fintech, education,...
  - Legal-specific LLMs: ChatLaw, LawGPT, SaulLM-7B....
- **Multi-agent Collaboration System for legal QA**
  - LawLuo, ChatLaw
- **Vietnamese Legal QA system**
  - ViGPT-Law

RELATED WORKS

Legal System Framework

- **Legal Data with Knowledge graph**

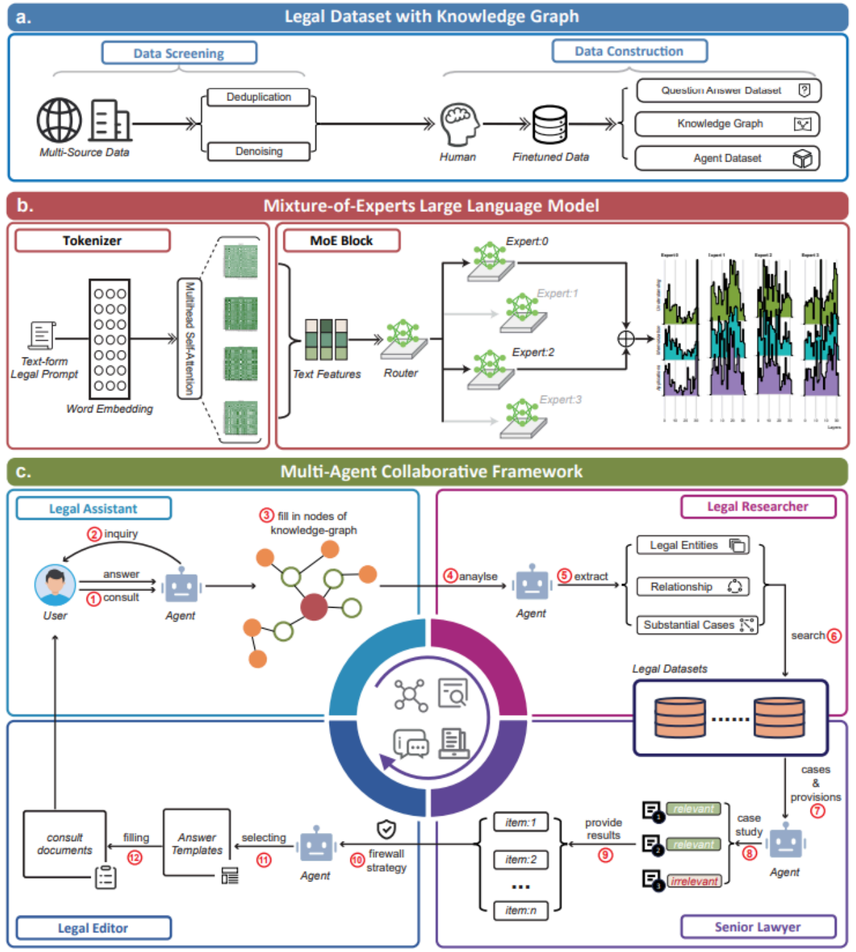
  Integrate multiple data source to construct Knowledge Graph

- **Mixture-of-expert LLM**

  A virtual legal team composed of multiple virtual legal experts specializing in different legal fields

- **Multi-agent Framework**

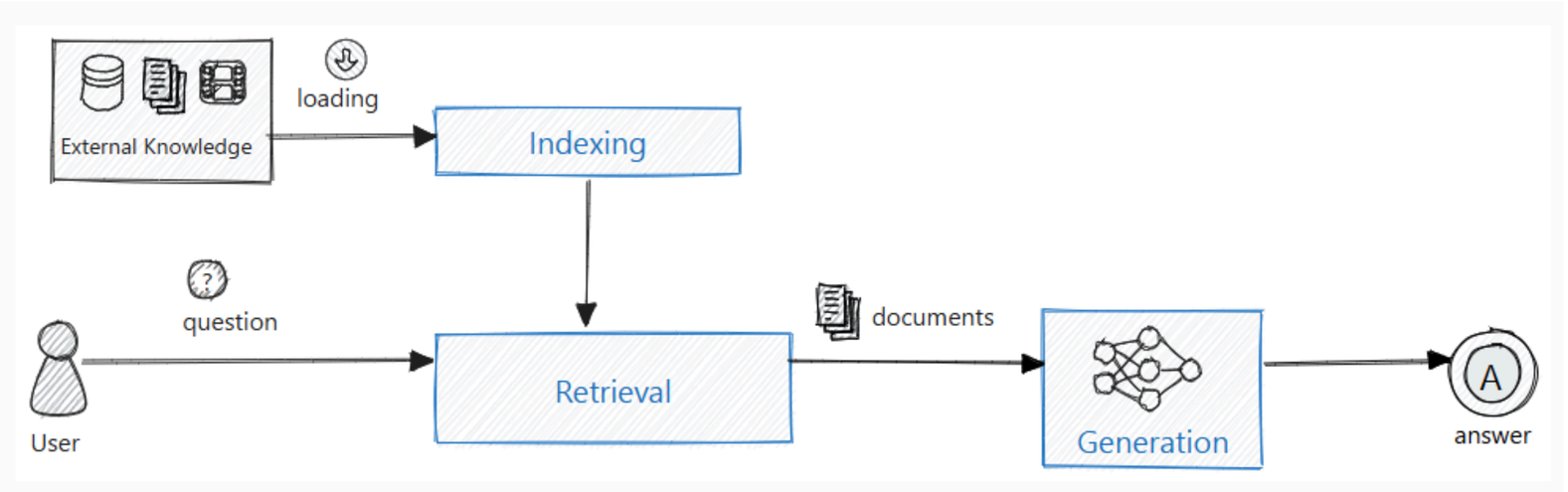
  Contains 4 agents: Legal Assistant, Legal Researcher, Senior Lawyer and Legal Editor



RELATED WORKS

Retrieval Augumented Generation(RAG)

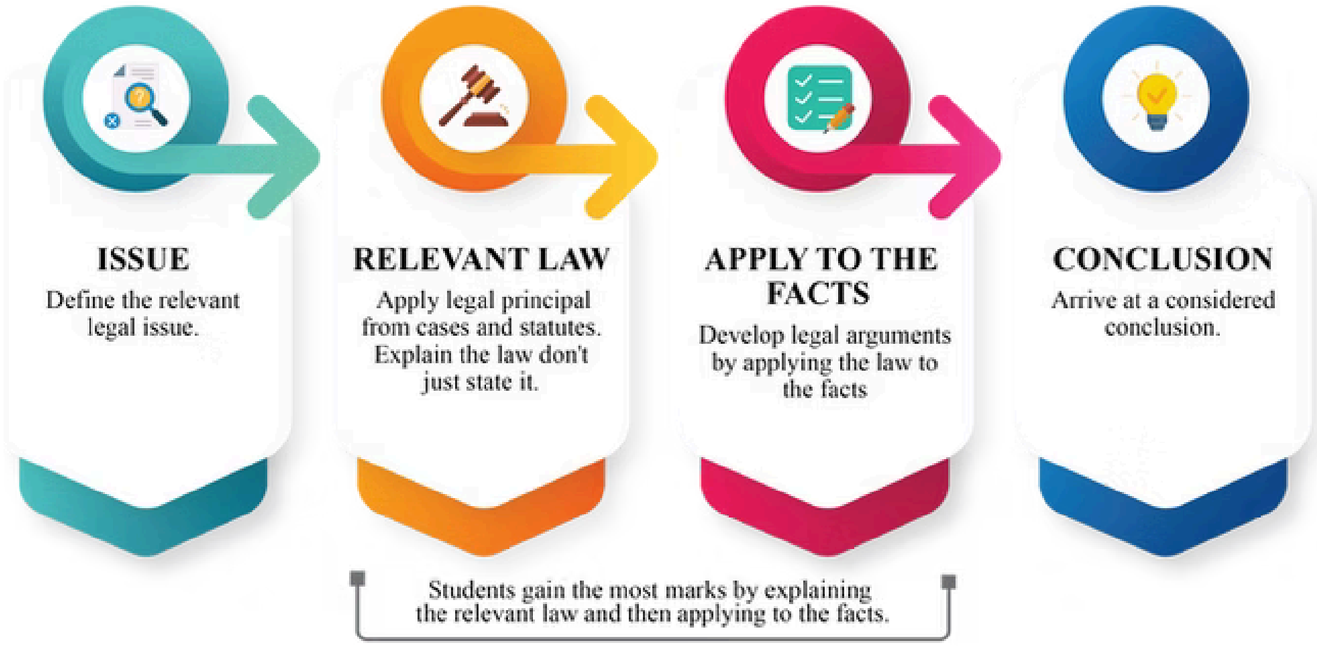
According to [], Retrieval-Augmented Generation (RAG) helps in legal practice by combining external legal databases with AI-generated content to provide more accurate and context-aware legal research and document drafting



RELATED WORKS

Legal Analysis

IRAC Method [] break down any legal scenario into these components which will help make your legal analysis and answers more organized. IRAC method is considered a highly helpful technique in legal analysis



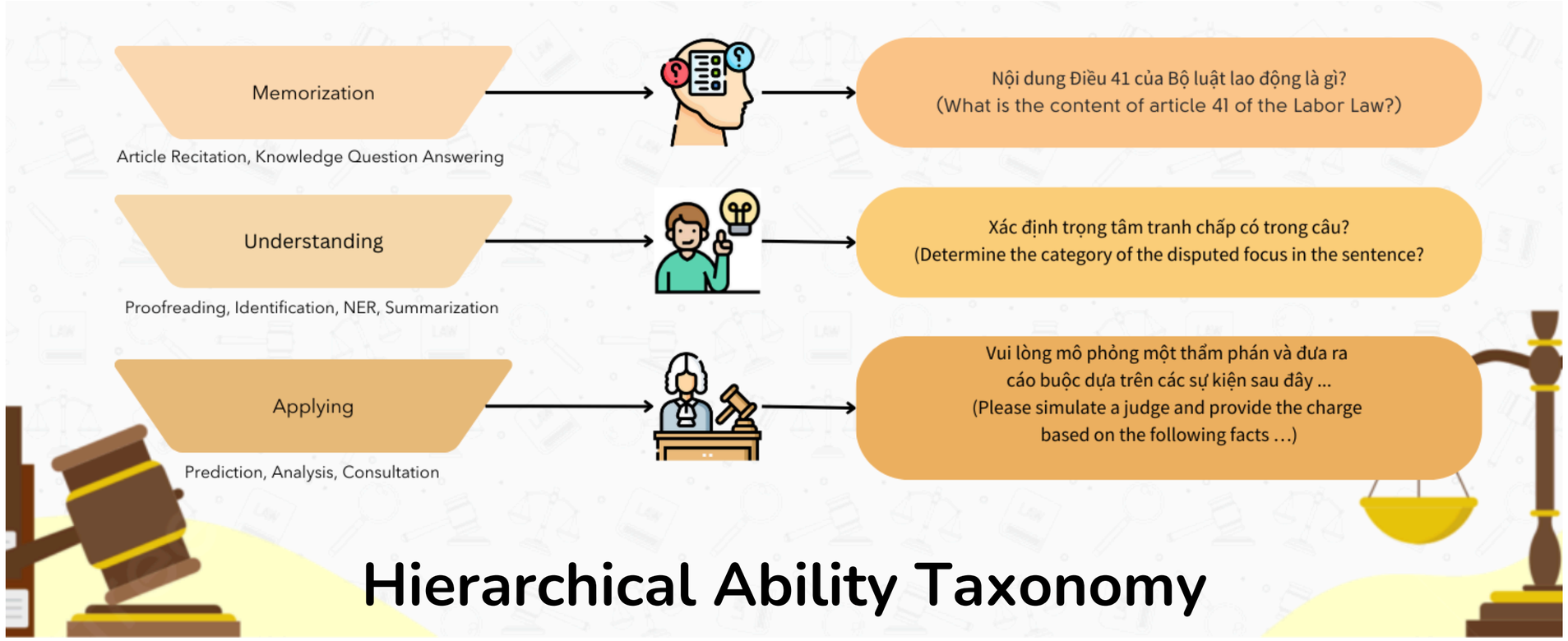
CONTRIBUTION

- Legal Benchmark and Evaluation for Vietnamese
- Fine-Tuned Large Language Model
- Multi-agent Collaboration for Vietnamese Legal Consultation

2 METHODOLOGY

METHODOLOGY

Vietnamese Legal Benchmark



Hierarchical Ability Taxonomy

METHODOLOGY

Data Source and Selected Tasks

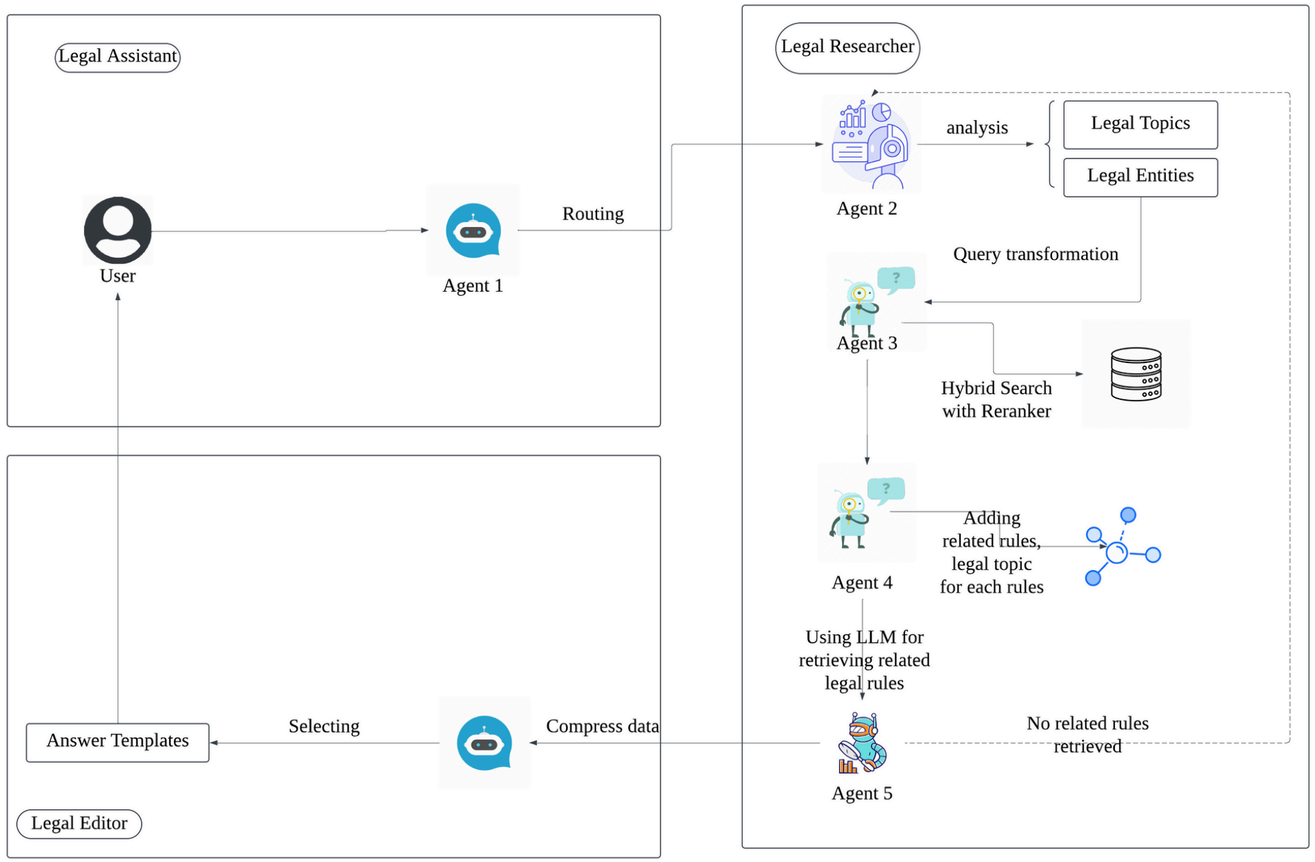
Cognitive Level	ID	Task	Data Source	Metric	Type
Legal Knowledge Memorization	1-1	Article Recitation	VLD[1]	Rouge-L	Generation
	1-2	Knowledge Question Answering	Vietjack[2]	Accuracy	SLC
Legal Knowledge Understanding	2-1	Document Proofreading	VLD[1]	GLEU Score	Generation
	2-2	Dispute Focus Identification	Congbobanan[3]	F1 score	MLC
	2-3	Marital Disputes Identification	Congbobanan[3]	F1 score	MLC
	2-4	Issue Topic Identification	Congbobanan[3]	Accuracy	SLC
	2-5	Reading Comprehension	Congbobanan[3]	rc-F1	Extraction
	2-6	Named-Entity Recognition	Congbobanan[3]	soft-F1	Extraction
	2-7	Opinion Summarization	PLC[4]	Rouge-L	Generation
	2-8	Event Detection	Thuvienphapluat[5]	F1 score	MLC
	2-9	Trigger Word Extraction	Thuvienphapluat[5]	soft-F1	Extraction
Legal Knowledge Applying	3-1	Fact-based Article Prediction	Congbobanan[3]	F1 score	MLC
	3-2	Scene-based Article Prediction	Thuvienphapluat	Rouge-L	Generation
	3-3	Charge Prediction	Congbobanan[3]	F1 score	MLC
	3-4	Prison Term Prediction w.o Article	Congbobanan[3]	nLog-distance	Regression
	3-5	Prison Term Prediction w. Article	Congbobanan[3]	nLog-distance	Regression
	3-6	Case Analysis	Vietjack[2]	Accuracy	SLC
	3-7	Consultation	MPS[6]	Rouge-L	Generation

Table 2.1: Task list in Vietnamese Legal Benchmark. Tasks correspond to cognitive dimensions: legal knowledge memorization, understanding and applying, and 5 task types: generation, single-label classification (SLC), multi-label classification (MLC), regression, and extraction.

METHODOLOGY

Multi-agent Framework

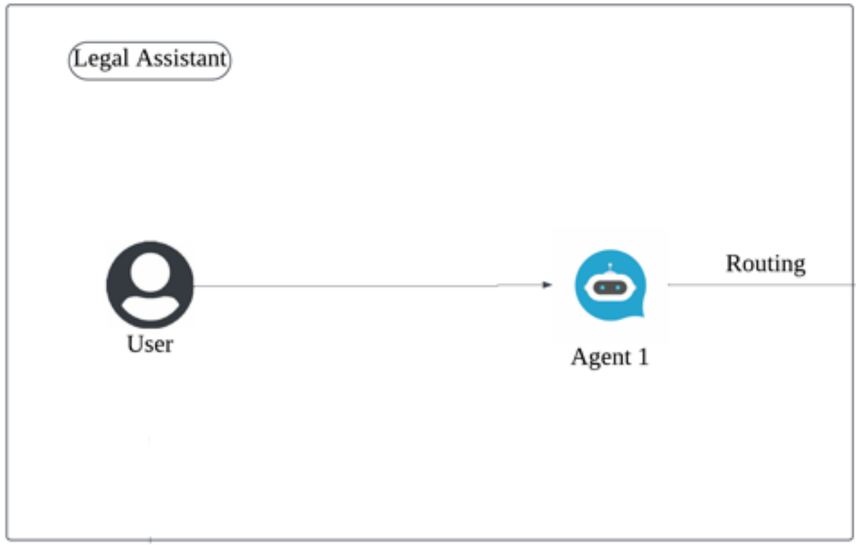
The workflow streamlines legal problem-solving by analyzing queries, retrieving precise legal rules, and providing clear, relevant answers, leveraging AI for efficient, context-aware legal assistance.



METHODOLOGY

Multi-agent Framework

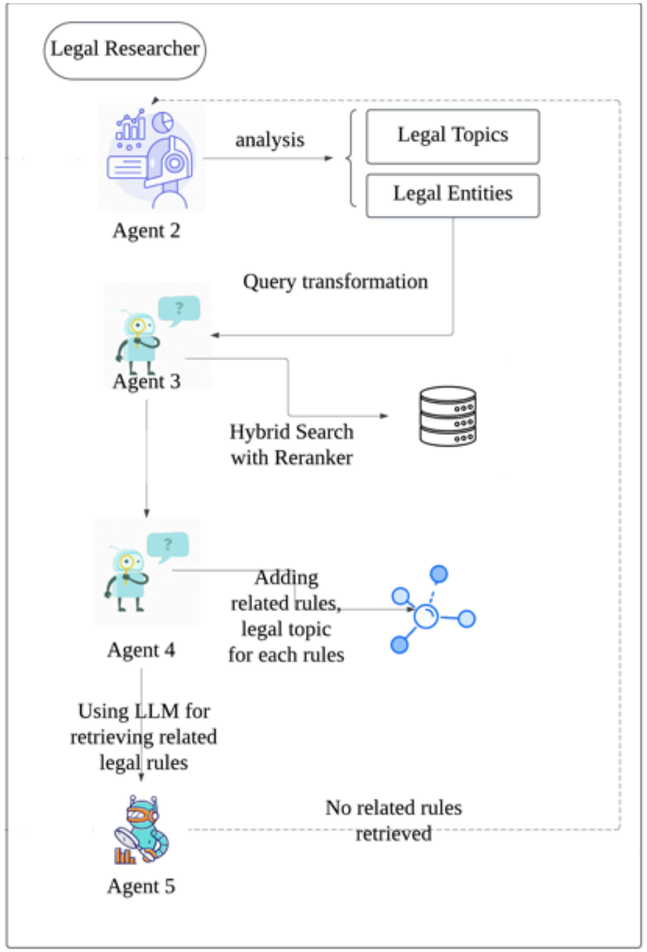
- The legal assistant in the workflow routes user queries, processes them through AI agents for analysis and retrieval, and delivers concise, relevant legal answers using structured templates, ensuring efficient support for legal



METHODOLOGY

Multi-agent Framework

- **Analysis**: Queries are analyzed, decomposed by complexity, and mapped to legal topics, entities, and relationships using topic extraction and metadata attachments.
- **Hybrid Search:** Combines dense vector search, keyword retrieval, and knowledge graph exploration to find relevant legal provisions, re-ranking results for precision and relational context.
- **Supervise & Recursive Retrieval**: Validates results with knowledge graphs, uses LLMs for checking and recursively refines retrieval to ensure comprehensive legal output.



METHODOLOGY

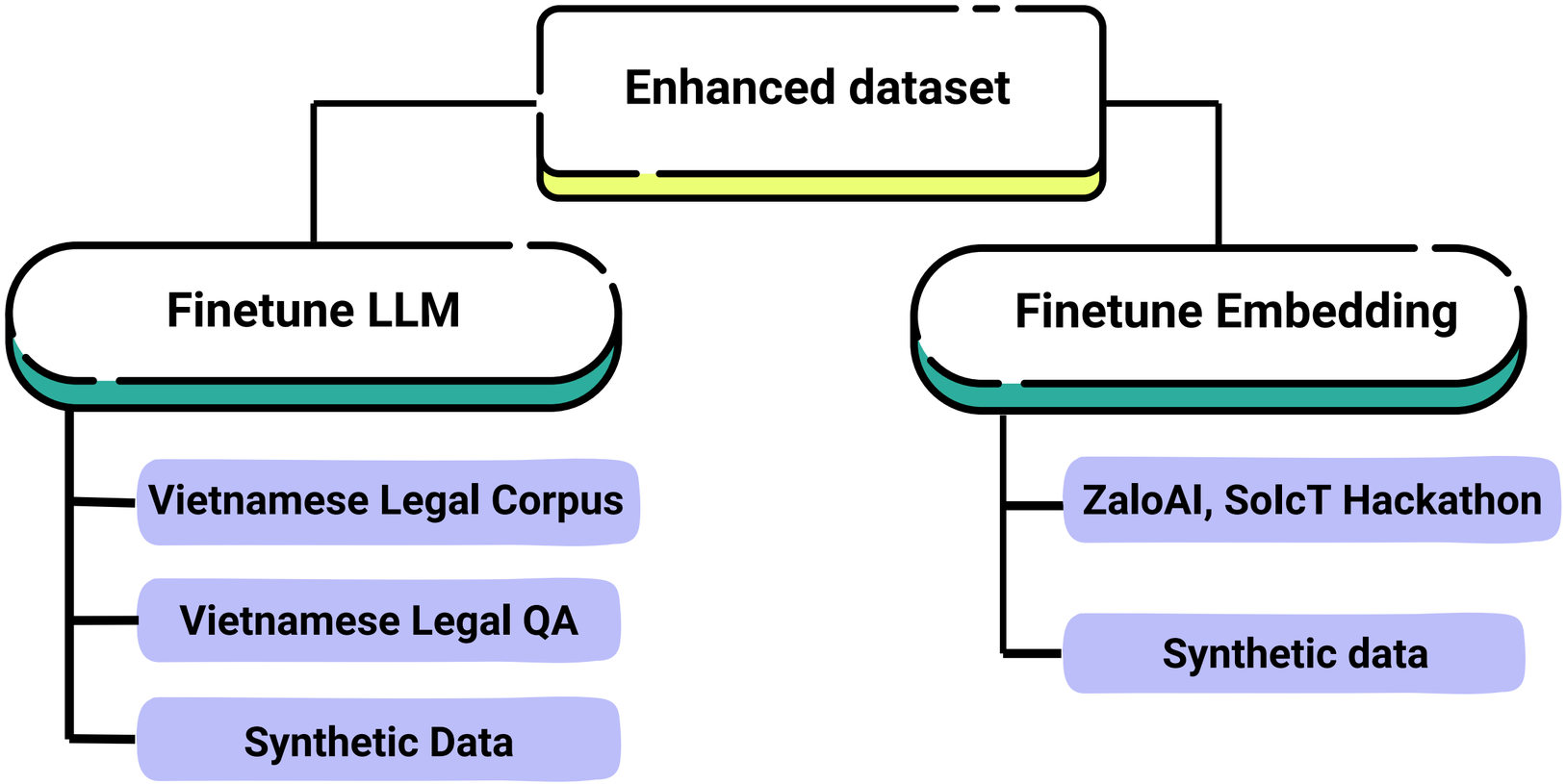
Multi-agent Framework

- **Legal Editor**: Selects templates based on query type, structures responses with retrieved legal rules and analyses, and ensures clarity and actionable alignment with the user's query.



METHODOLOGY

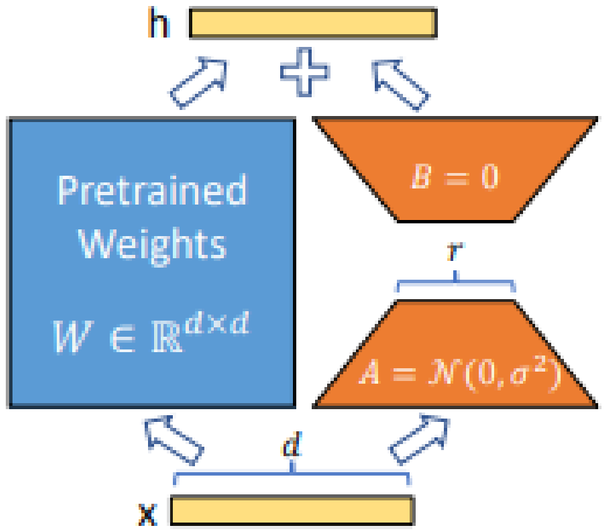
Enhanced Dataset for finetuning



METHODOLOGY

Instruction Finetuning LLM

- Fine-tuning adapts a pre-trained model to a specific domain by training only select parameters, reducing computational costs and preserving previously learned knowledge through techniques like LoRA.

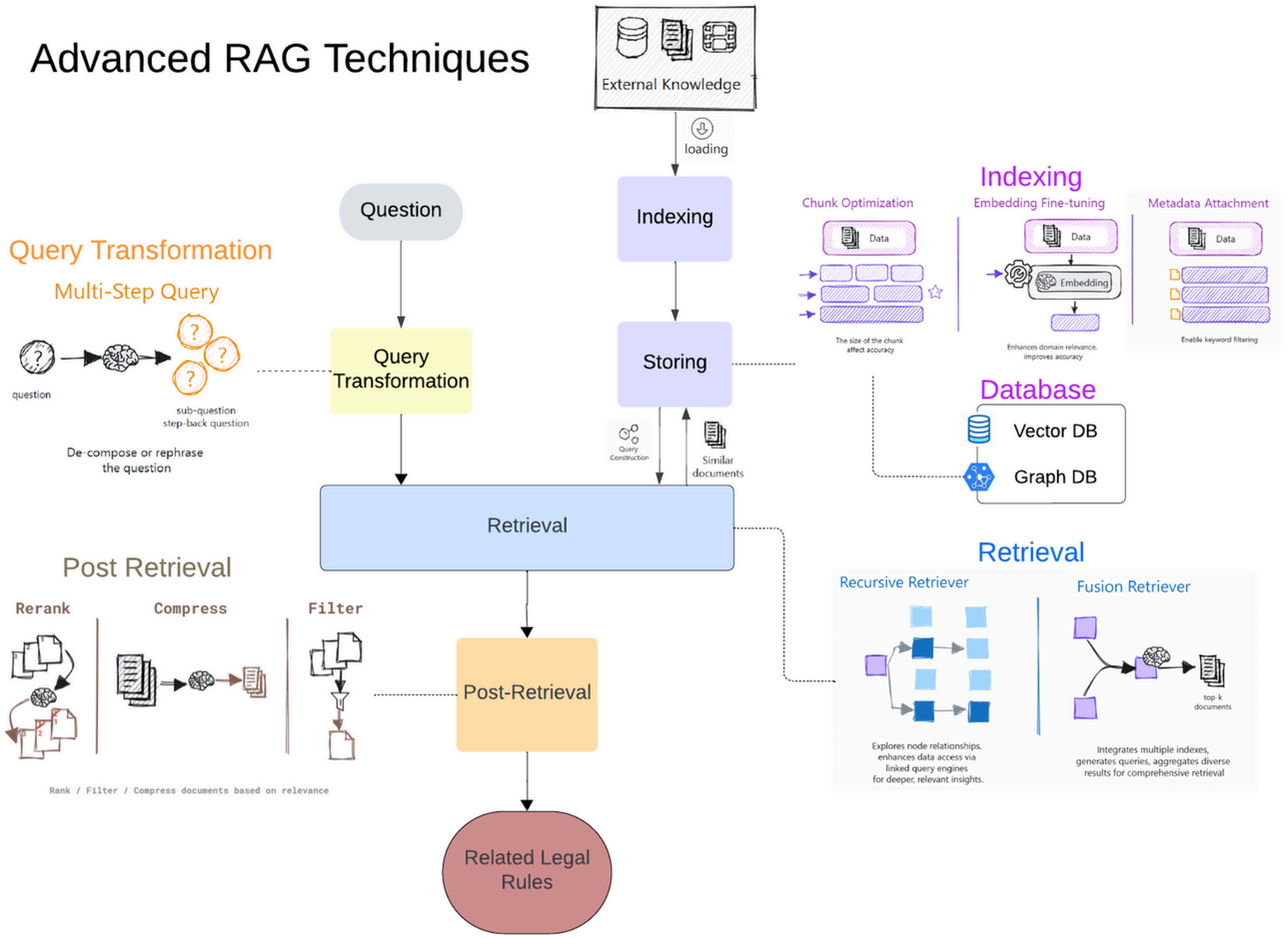


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METHODOLOGY

Advanced RAG

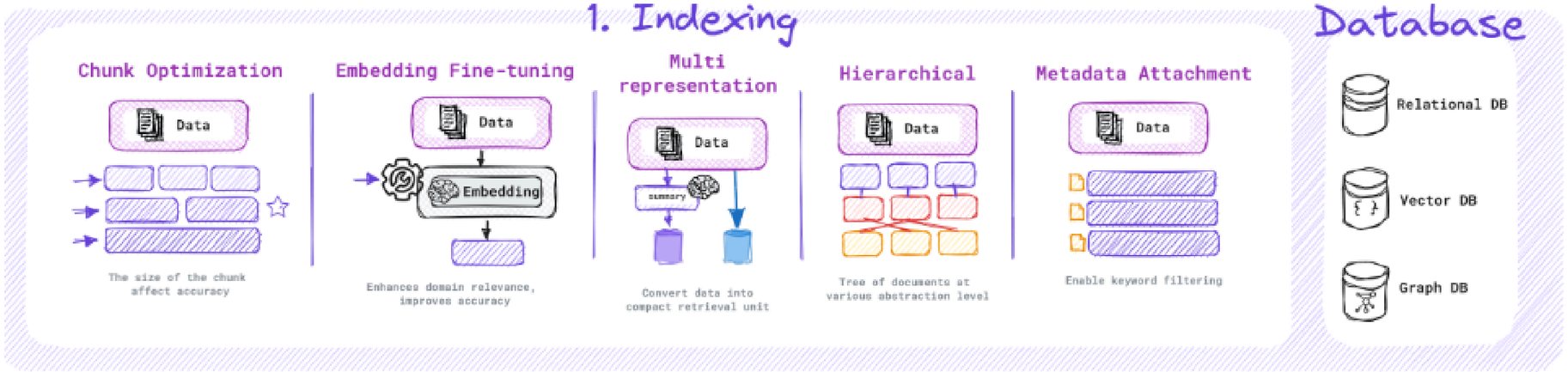
Advanced flow for (RAG) techniques, highlighting processes to optimize query handling, retrieval, and post-retrieval operations



METHODOLOGY

Advanced RAG

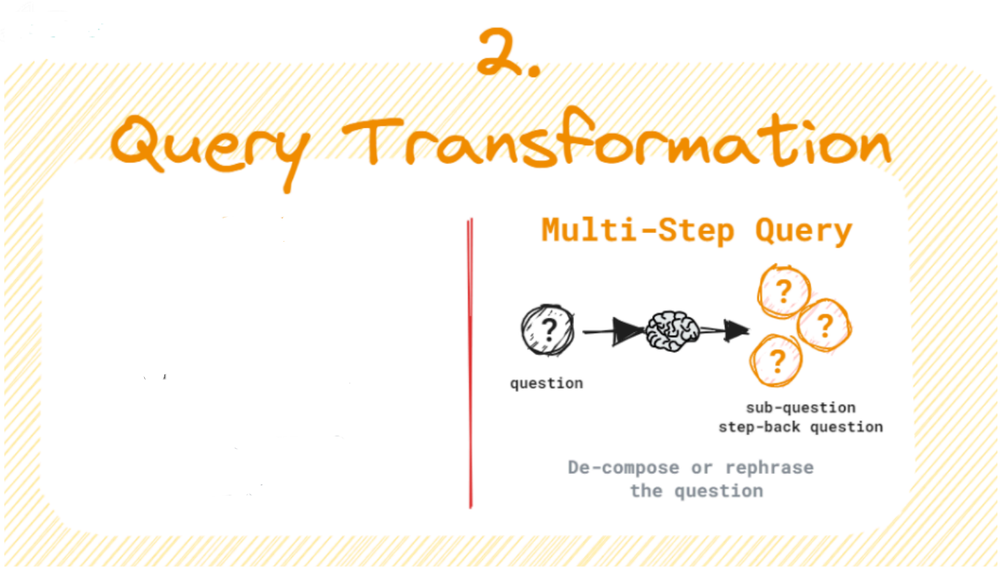
- Construct legal data by organizing the legal corpus into thematic sections, articles, and clauses, enriched with semantic metadata for hierarchical structure and precise retrieval of legal texts.
- Optimize retrieval with chunking, embedding fine-tuning, and metadata integration, ensuring context-rich and efficient access to legal provisions through vector databases and knowledge graphs.



METHODOLOGY

Advanced RAG

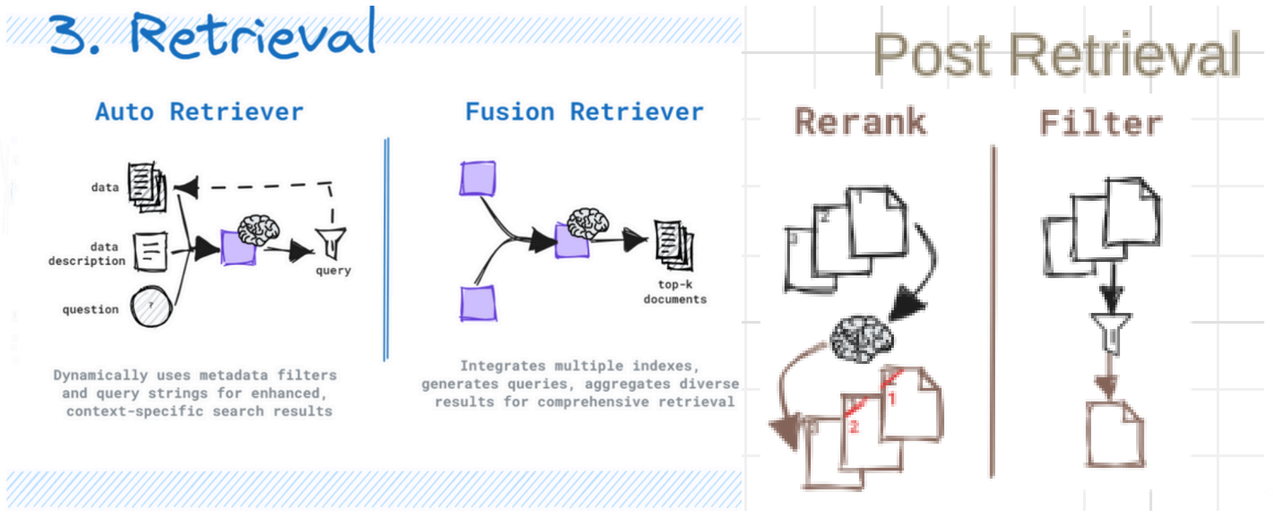
- Query Transformation refines user inputs by rephrasing ambiguous queries and decomposing complex ones, enhancing alignment with indexed legal data and improving precision in retrieving relevant legal provisions.



METHODOLOGY

Advanced RAG

- Hybrid Retrieval combines vector-based and graph-based approaches, refining semantic alignment via re-ranking and leveraging relational insights from knowledge graphs for comprehensive, context-aware retrieval of legal provisions.



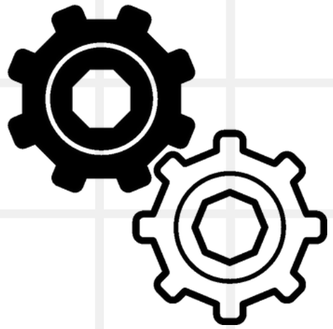
3

EXPERIMENT

Metrics

Cognitive Level	ID	Task	Data Source	Metric	Type
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	2-7	Opinion Summarization	PLO[4]	Rouge-L	Generation
	2-8	Event Detection	Thuvienphapluat[5]	F1 score	MLC
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	3-4	Prison Term Prediction w.o Article	Congbobanan[3]	nLog-distance	Regression
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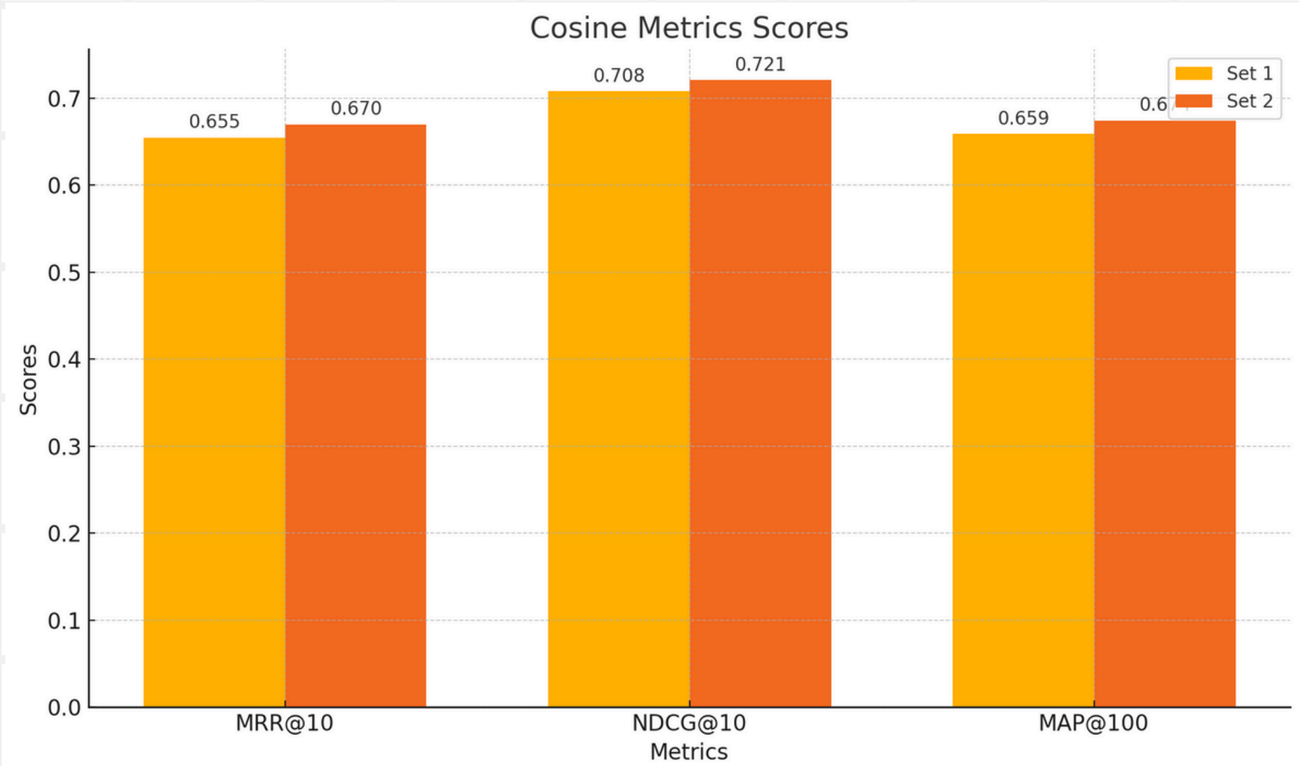
EXPRIMENT SETTINGS

Environment

- **Pytorch & Cuda 12**
- **Python 3.9**
- **Anaconda 3**
- **GPU NVIDIA GTX 3090**

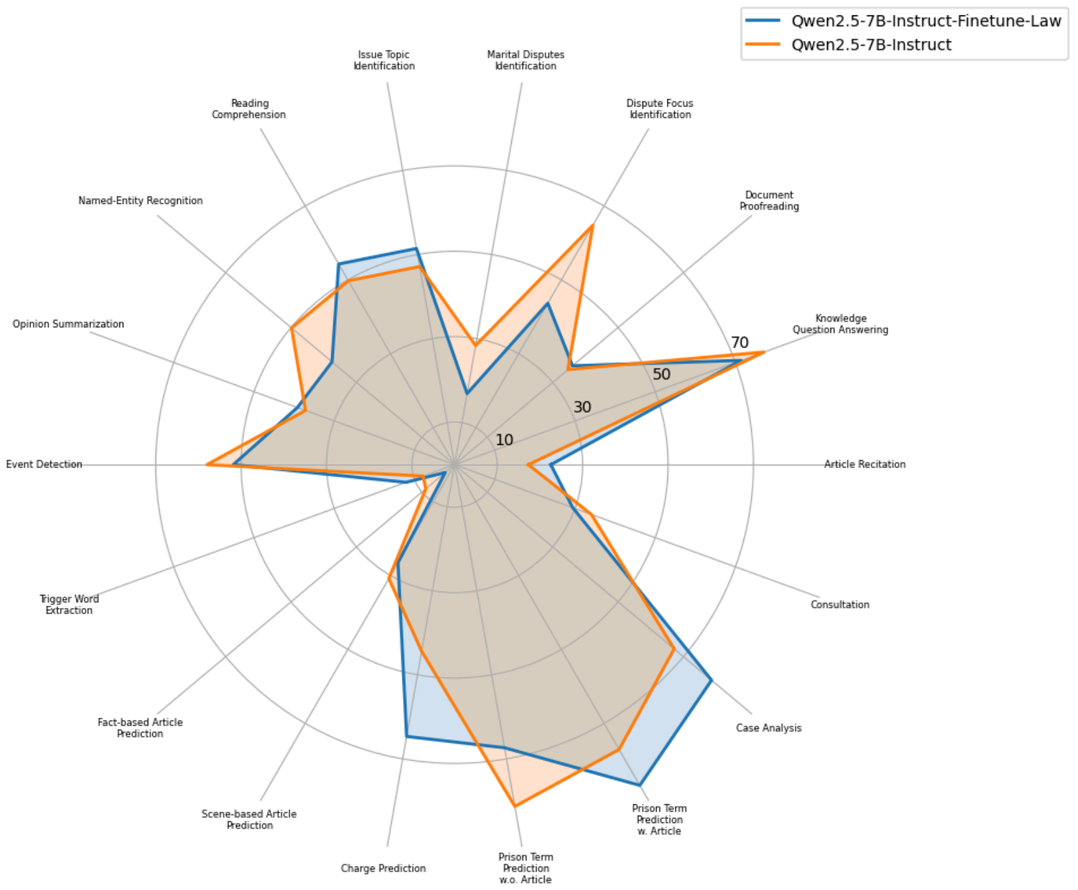
Hyper parameter	Value
base lr	5.00E-06
batch size	4
num_epoch	1
optimizer	adamw
weight decay	0.00001

Result



Embedding base vs Finetune

Result



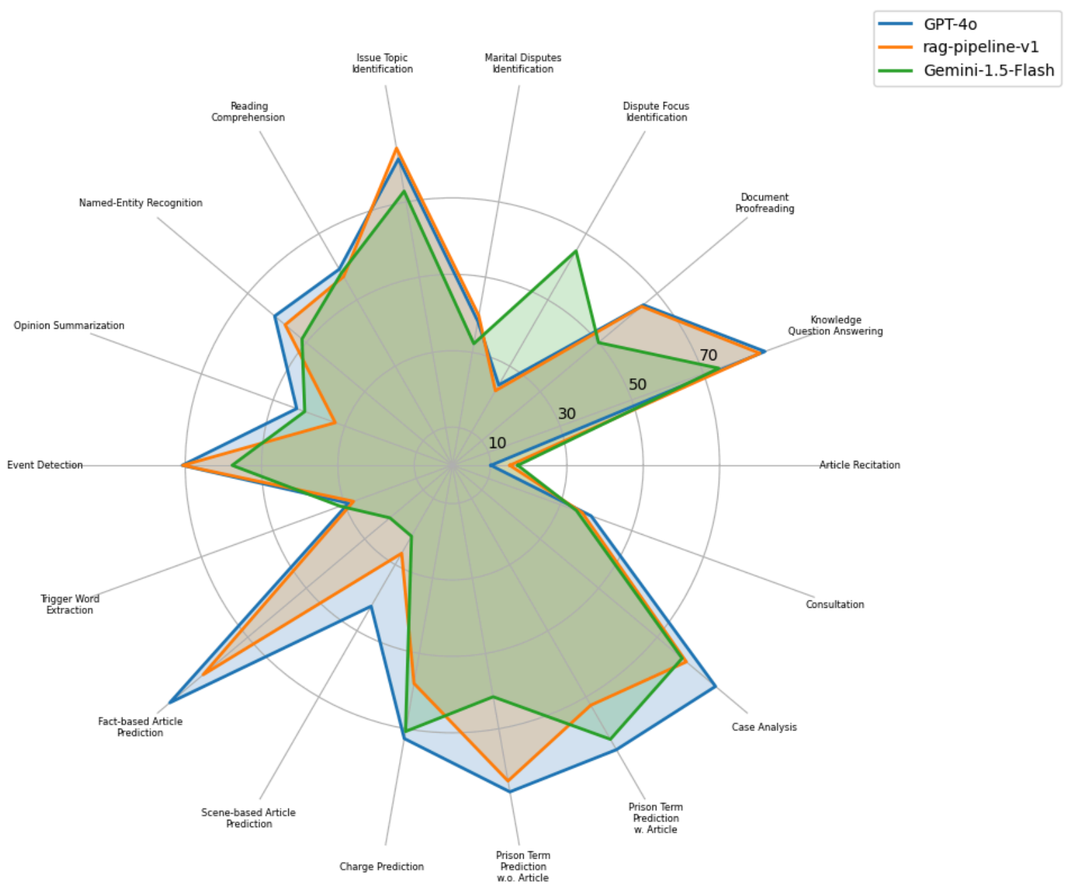
Comparison between Model base vs Model Fine tune

Result

Category	Multilingual LLMs			Vietnamese Oriented LLMs		Law Specific LLMs
model_name	GPT-4o	Gemini-1.5-Flash	Qwen2.5-7B-Instruct	Ml4uLLM-7B-Chat	Vistral-7B-Chat	Qwen2.5-7B-Instruct-Finetune-Law
task						
1-1	10.09	17.12	17.29	25.13	25.65	22.54
1-2	87.14	74.29	77.14	14.29	30.00	71.43
2-1	65.32	49.97	34.65	31.87	37.04	36.06
2-2	24.29	64.79	64.79	59.15	35.21	43.66
2-3	38.31	32.34	28.36	12.44	0.00	16.92
2-4	81.43	72.86	47.14	42.86	0.00	51.43
2-5	59.27	58.13	49.75	59.01	56.99	54.29
2-6	60.77	51.44	49.86	42.77	32.20	37.47
2-7	43.42	41.18	37.15	40.37	44.02	39.17
2-9	70.78	57.70	57.98	25.33	19.13	51.84
2-10	28.98	31.27	7.83	11.40	7.71	12.09
3-1	96.67	21.38	8.81	74.02	88.81	2.86
3-2	42.58	21.43	30.80	35.22	34.27	26.53
3-3	72.61	70.73	44.39	52.34	63.19	64.66
3-4	86.78	61.51	81.31	86.18	88.89	67.36
3-5	85.93	82.79	77.07	76.81	79.44	86.77
3-6	90.00	78.57	67.14	7.14	47.14	78.57
3-7	38.67	34.57	34.10	36.53	35.94	29.45
Average	60.17	51.23	45.31	40.71	40.31	44.06

Table : Model performance of selected LLMs

Result



Comparison between Our system vs GPT4o vs gemini1.5 flash

4

CONCLUSION & FUTURE WORKS

CONCLUSION

- This thesis proposes Vietnamese Legal Benchmark multi-agent system in legal consultation using knowledge graph and LLMs
- Our proprietary framework enhanced reasoning accuracy and reduced hallucinations effectively.

FUTURE WORKS

- Improve the results Vietnamese Legal Benchmark
- Fine tune LLMs with other methods, such as DPO
- Build a legal consultation application to help users experience and get the fastest and most accurate legal advice

REFERENCE

5 DEMO

THANK YOU